

```
!pip install pymongo
```

```
Collecting pymongo
```

```
  Downloading pymongo-4.15.5-cp312-cp312-  
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.m  
etadata (22 kB)
```

```
Collecting dnspython<3.0.0,>=1.16.0 (from pymongo)
```

```
  Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
```

```
Downloaded pymongo-4.15.5-cp312-cp312-  
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl  
(1.7 MB)
```

```
0.0/1.7 MB ? eta -:-:--
```

```
1.7/1.7 MB 76.8 MB/s eta
```

```
0:00:00
```

```
0.0/331.1 kB ? eta -:-:--
```

```
331.1/331.1 kB 33.3 MB/s eta
```

```
0:00:00
```

```
ongo
```

```
Successfully installed dnspython-2.8.0 pymongo-4.15.5
```

```
!pip install pinecone
```

```
Collecting pinecone
```

```
  Downloading pinecone-8.0.0-py3-none-any.whl.metadata (11 kB)
```

```
Requirement already satisfied: certifi>=2019.11.17 in  
/usr/local/lib/python3.12/dist-packages (from pinecone) (2025.11.12)
```

```
Requirement already satisfied: orjson>=3.0.0 in
```

```
/usr/local/lib/python3.12/dist-packages (from pinecone) (3.11.5)
```

```
Collecting pinecone-plugin-assistant<4.0.0,>=3.0.1 (from pinecone)
```

```
  Downloading pinecone_plugin_assistant-3.0.1-py3-none-  
any.whl.metadata (30 kB)
```

```
Requirement already satisfied: pinecone-plugin-interface<0.1.0,>=0.0.7  
in /usr/local/lib/python3.12/dist-packages (from pinecone) (0.0.7)
```

```
Requirement already satisfied: python-dateutil>=2.5.3 in
```

```
/usr/local/lib/python3.12/dist-packages (from pinecone) (2.9.0.post0)
```

```
Requirement already satisfied: typing-extensions>=3.7.4 in
```

```
/usr/local/lib/python3.12/dist-packages (from pinecone) (4.15.0)
```

```
Requirement already satisfied: urllib3>=1.26.5 in
```

```
/usr/local/lib/python3.12/dist-packages (from pinecone) (2.5.0)
```

```
Collecting packaging<25.0,>=24.2 (from pinecone-plugin-  
assistant<4.0.0,>=3.0.1->pinecone)
```

```
  Downloading packaging-24.2-py3-none-any.whl.metadata (3.2 kB)
```

```
Requirement already satisfied: requests<3.0.0,>=2.32.3 in
```

```
/usr/local/lib/python3.12/dist-packages (from pinecone-plugin-  
assistant<4.0.0,>=3.0.1->pinecone) (2.32.4)
```

```
Requirement already satisfied: six>=1.5 in
```

```
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.5.3-  
>pinecone) (1.17.0)
```

```
Requirement already satisfied: charset_normalizer<4,>=2 in
```

```
/usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.3-
```

```

>pinecone-plugin-assistant<4.0.0,>=3.0.1->pinecone) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.3-
>pinecone-plugin-assistant<4.0.0,>=3.0.1->pinecone) (3.11)
Downloading pinecone-8.0.0-py3-none-any.whl (745 kB)
_____ 745.9/745.9 kB 53.9 MB/s eta
0:00:00
_____ 280.9/280.9 kB 29.3 MB/s eta
0:00:00
_____ 65.5/65.5 kB 6.9 MB/s eta
0:00:00
pting uninstall: packaging
  Found existing installation: packaging 25.0
  Uninstalling packaging-25.0:
    Successfully uninstalled packaging-25.0
Successfully installed packaging-24.2 pinecone-8.0.0 pinecone-plugin-
assistant-3.0.1

{"id":"06698edc3c154e81b71fcfc16a05a665","pip_warning":{"packages":
["packaging"]}}
```

!pip install transformers

```

Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (4.57.3)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from transformers) (3.20.0)
Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.36.0)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (25.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: requests in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.32.4)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.1)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.1)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.34.0->transformers) (2025.3.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in

```

```

/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.34.0->transformers) (4.15.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.34.0->transformers) (1.2.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers)
(3.4.4)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers)
(3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers)
(2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers)
(2025.11.12)

from pymongo.mongo_client import MongoClient
from pymongo.server_api import ServerApi

uri =
"mongodb+srv://divyanshuchauhan471:divyanshu7123@cluster1.ifhfhqv.mong
odb.net/?appName=Cluster1"

# Create a new client and connect to the server
client = MongoClient(uri, server_api=ServerApi('1'))

# Send a ping to confirm a successful connection
try:
    client.admin.command('ping')
    print("Pinged your deployment. You successfully connected to
MongoDB!")
except Exception as e:
    print(e)

Pinged your deployment. You successfully connected to MongoDB!

from pinecone import Pinecone
PINECONE_API_KEY="add your actual Pinecone API key here"
pc = Pinecone(api_key=PINECONE_API_KEY)
index = pc.Index("divyanshu")

index
<pinecone.db_data.index.Index at 0x7ef9d976bc20>

db = client['divyanshu']
collection = db['divyanshucollection']

```

```
!pip install sentence_transformers
```

```
Requirement already satisfied: sentence_transformers in  
/usr/local/lib/python3.12/dist-packages (5.2.0)  
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in  
/usr/local/lib/python3.12/dist-packages (from sentence_transformers)  
(4.57.3)  
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-  
packages (from sentence_transformers) (4.67.1)  
Requirement already satisfied: torch>=1.11.0 in  
/usr/local/lib/python3.12/dist-packages (from sentence_transformers)  
(2.9.0+cu126)  
Requirement already satisfied: scikit-learn in  
/usr/local/lib/python3.12/dist-packages (from sentence_transformers)  
(1.6.1)  
Requirement already satisfied: scipy in  
/usr/local/lib/python3.12/dist-packages (from sentence_transformers)  
(1.16.3)  
Requirement already satisfied: huggingface-hub>=0.20.0 in  
/usr/local/lib/python3.12/dist-packages (from sentence_transformers)  
(0.36.0)  
Requirement already satisfied: typing_extensions>=4.5.0 in  
/usr/local/lib/python3.12/dist-packages (from sentence_transformers)  
(4.15.0)  
Requirement already satisfied: filelock in  
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-  
>sentence_transformers) (3.20.0)  
Requirement already satisfied: fsspec>=2023.5.0 in  
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-  
>sentence_transformers) (2025.3.0)  
Requirement already satisfied: packaging>=20.9 in  
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-  
>sentence_transformers) (24.2)  
Requirement already satisfied: pyyaml>=5.1 in  
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-  
>sentence_transformers) (6.0.3)  
Requirement already satisfied: requests in  
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-  
>sentence_transformers) (2.32.4)  
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in  
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-  
>sentence_transformers) (1.2.0)  
Requirement already satisfied: setuptools in  
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-  
>sentence_transformers) (75.2.0)  
Requirement already satisfied: sympy>=1.13.3 in  
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-  
>sentence_transformers) (1.14.0)  
Requirement already satisfied: networkx>=2.5.1 in  
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
```

```
>sentence_transformers) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (3.1.6)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (12.6.77)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (12.6.77)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (12.6.80)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (12.6.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (11.3.0.4)
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (10.3.7.77)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (11.7.1.2)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (12.5.4.2)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.3.20 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (3.3.20)
Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (12.6.77)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (12.6.85)
Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (1.11.1.6)
Requirement already satisfied: triton==3.5.0 in
```

```

/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0-
>sentence_transformers) (3.5.0)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence_transformers) (2.0.2)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence_transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence_transformers) (0.22.1)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence_transformers) (0.7.0)
Requirement already satisfied: joblib>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn-
>sentence_transformers) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn-
>sentence_transformers) (3.6.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch>=1.11.0->sentence_transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0-
>sentence_transformers) (3.0.3)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->huggingface-
hub>=0.20.0->sentence_transformers) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests->huggingface-
hub>=0.20.0->sentence_transformers) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests->huggingface-
hub>=0.20.0->sentence_transformers) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->huggingface-
hub>=0.20.0->sentence_transformers) (2025.11.12)

```

```

from sentence_transformers import SentenceTransformer
embedding_model = SentenceTransformer("thenlper/gte-large")

```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to

```

```
access public models or datasets.  
warnings.warn(  

```

```
{"model_id":"d6023ac97b0d42698386ea63bf7e8459","version_major":2,"version_minor":0}
```

```
{"model_id":"be52abec065749a5a6173cc8b9dee97b","version_major":2,"version_minor":0}
```

```
{"model_id":"b3e6de65cea341a98b240d64d8cdc821","version_major":2,"version_minor":0}
```

```
{"model_id":"cfc6444e3dae45c4b7311e65af85d962","version_major":2,"version_minor":0}
```

```
{"model_id":"46c2c5c709514769ab0f6cd470df5d86","version_major":2,"version_minor":0}
```

```
{"model_id":"edc16ccad5cb468a99791a5da1cd6dca","version_major":2,"version_minor":0}
```

```
{"model_id":"8e019b2f7f9045219cbb9349524b2928","version_major":2,"version_minor":0}
```

```
{"model_id":"28a708c38f894816aeedc69c7df633ba","version_major":2,"version_minor":0}
```

```
{"model_id":"4cbb2cc45c4c4126b091ac4221e1410a","version_major":2,"version_minor":0}
```

```
{"model_id":"170f916f0ea24f20829ae4bb9a1817c8","version_major":2,"version_minor":0}
```

```
cursor = collection.watch(full_document='updateLookup')
```

```
print("Change stream is now open.")
```

```
while True:
```

```
    change = next(cursor)
```

```
    if change['operationType'] == 'insert':
```

```
        document = change['fullDocument']
```

```
        vector = embedding_model.encode(document['fullplot'])
```

```
        vector = vector.tolist()
```

```
        vector = [float(x) for x in vector]
```

```
        upsert_data = (str(document['_id']), vector)
```

```
        index.upsert([upsert_data])
```

```
    elif change['operationType'] == 'update':
```

```
        document = change['fullDocument']
```

```
        document_id = document['_id']
```

```
        updated_fields = change['updateDescription']['updatedFields']
```

```
        if updated_fields.get('fullplot'):
```

```
        vector = embedding_model.encode(updated_fields['fullplot'])
        upsert_data = (str(document_id), vector)
        index.upsert([upsert_data])
    elif change['operationType'] == 'delete':
        index.delete(ids=[str(change['documentKey']['_id'])])
```

Change stream is now open.